ORBIT VIRTUAL PATIENT SIMULATOR

Biophysical Foundations & Mathematical Derivations Monograph

A Continuous Mathematical Compendium of Hemodynamics, Electrophysiology, Fluid Mechanics, Bioacoustics, and WebGL Systems

Author / Architecture: Orbit Medical Engineering Group
Target Curriculum: NMC CBME, Harrison's, Guyton & Hall, Braunwald
Monograph Reference: ORBIT-BIO-MATH-2026-V1
Mathematical Class: Multi-Scale Coupled ODEs & PDEs
Execution Kernel: 100 Hz Modified Nodal Analysis
Classification: High-Fidelity Clinical Simulator

1. Executive Preamble: Continuous First-Principles Modeling

Contemporary medical simulators frequently rely on discrete state-machine heuristics or coarse rule-based lookups (e.g., if blood loss > 1000 mL, reduce systolic blood pressure by 25 mmHg). While computationally trivial, such empirical approximations break down during complex multi-organ pathology, drug-disease interactions, and dynamic resuscitation. The **Orbit Virtual Patient Simulator** rejects phenomenological shortcuts in favor of **continuous, closed-loop biophysical models grounded in classical conservation laws**.

Every physiological variable in Orbit—from left ventricular pressure trajectories to 12-lead surface ECG voltages, sub-valvular acoustic shear flutter, and trans-microvascular glycocalyx fluid exchange—arises deterministically from coupled ordinary and partial differential equations (ODEs/PDEs). This document provides the formal mathematical derivations, mechanical analogies, dimensional validations, and primary literature citations supporting Orbit's engine.

2. Cardiovascular Hemodynamics & Ventricular Mechanics

2.1. The Wiggers Diagram & Cardiac Cycle Phase Mechanics

The cardiac cycle represents the periodic conversion of biochemical cross-bridge potential energy into mechanical stroke work and hydraulic arterial pressure. Carl J. Wiggers (1915) synthesized the temporal relationship between intracardiac pressures, ventricular volumes, electrograms, and heart sounds into seven distinct mechanical phases:

Phase	Mechanical State	A-V Valves	SL Valves	Governing Dynamics
1. Atrial Systole	Active ventricular topping-up (~15-20% EDV)	Open	Closed	Atrial booster pump; a-wave on CVP; S4 in stiff LV
2. Isovolumetric Contraction	Rapid dP/dt pressure spike; volume fixed	Closed (M1/T1)	Closed	P_LV rises 10 -> 80 mmHg; S1 sound generated
3. Rapid Ejection	Accelerating aortic inflow; max LV pressure	Closed	Open	Inertial blood acceleration; peak systolic pressure
4. Reduced Ejection	Decelerating inflow; ventricular repolarization	Closed	Open	Aortic pressure exceeds LV; run-off into periphery

5. Isovolumetric Relaxation	Exponential pressure decay (-dP/dt); volume fixed	Closed	Closed (A2/P2)	Dicrotic notch / incisura; S2 sound generated
6. Rapid Ventricular Filling	Passive suction filling; rapid volume influx	Open	Closed	Elastic recoil suction; S3 gallop if volume overload
7. Diastasis (Reduced Filling)	Passive venous conduit flow; slow filling	Open	Closed	Filling rate asymptotically approaches zero

2.2. Time-Varying Ventricular Elastance Model: E(t)

To model ventricular pump action without imposing an unphysiological constant-pressure or constant-flow source, Orbit implements the seminal **time-varying elastance formulation** established by Hiroyuki Suga and Kiichi Sagawa (1974). The instantaneous pressure generated within the left or right ventricular cavity is expressed as a dynamic linear elastance operator acting upon the volume above a stress-free unstressed volume V_0:

$$P_{lv}(t) = E(t) \cdot [V_{lv}(t) - V_0]$$

Where E(t) [mmHg/mL] is instantaneous chamber elastance, and V_0 [mL] is unstressed cavity volume.

The elastance function $E(t)$ shifts periodically between a low passive diastolic compliance E_{min} (or E_d) and a high active inotropic peak E_{max} (or E_{es}, end-systolic elastance). Normalized time-varying elastance $E_N(t_N)$ displays a remarkable invariance across inotropic states, heart rates, and species, parameterized via the Double-Hill activation equation:

$$E_N(t_N) = 1.025 \cdot \left[\frac{(t_N / 0.303)^{1.32}}{1 + (t_N / 0.303)^{1.32}} \right] \cdot \left[\frac{1}{1 + (t_N / 0.508)^{21.9}} \right]$$

Double-Hill normalized elastance operator where t_N = t / T_max is normalized cycle time.

Physiological Significance: When contractility increases (e.g., inotropic stimulation with Dobutamine or Norepinephrine), E_{es} rotates counter-clockwise with a steeper slope, elevating stroke volume and peak systolic pressure without altering V_0. Conversely, cardiogenic shock and ischemic heart failure flatten E_{es}, collapsing stroke volume and elevating end-systolic volume.

2.3. End-Diastolic Pressure-Volume Relationship (EDPVR) & Frank-Starling Law

During diastole, the myocardium behaves as a nonlinear, viscoelastic hyperelastic tissue. As sarcomeres stretch, titin filaments and the extracellular collagen matrix resist further distension, yielding an exponential relationship:

$$P_{ed}(V) = P_0 \cdot \left(e^{\lambda \cdot (V - V_0)} - 1 \right)$$

Where P_0 is passive pressure scale factor, and \lambda [mL^{-1}] is myocardial stiffness coefficient.

The Frank-Starling Mechanism: Increased venous return augments end-diastolic volume (preload), stretching myocytes closer to optimal sarcomere overlap ($L_0 \approx 2.2\,\mu\text{m}$). This enhances troponin C affinity for Ca^{2+}, automatically magnifying contractile force on the subsequent beat without neural input.

2.4. Westerhof 4-Element Windkessel Arterial Model

Orbit models the systemic arterial tree via the 4-element Windkessel circuit (Westerhof et al., 1971), accounting for characteristic aortic impedance (R_c), total arterial compliance (C), peripheral resistance (R_p), and blood column inertance (L):

$$I(t) = C \frac{dP_{ao}}{dt} + \frac{P_{ao} - P_{ven}}{R_p} + \frac{1}{L} \int (P_{ao} - P_{art}) dt$$

Governing differential equation of central aortic pressure and pulsatile vascular impedance.

3. Vascular Fluid Dynamics & Hemodynamic Resistance

3.1. Incompressible Navier-Stokes Equations for Arterial Blood Flow

Blood is modeled as an incompressible Newtonian fluid (or Casson fluid in microvessels) obeying the Navier-Stokes momentum conservation equations:

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} \right) = -\nabla p + \mu \nabla^2 \mathbf{u} + \mathbf{f}$$

Where $\rho \approx 1060\,\text{kg/m}^3$, $\mu \approx 0.0035\,\text{Pa}\cdot\text{s}$ (blood dynamic viscosity), and p is hydraulic pressure.

3.2. Hagen-Poiseuille Resistance Derivation & Vessel Caliber Scaling

Jean Léonard Marie Poiseuille (1840) derived the analytical solution for laminar steady flow through a cylindrical conduit of radius r and length L. By integrating viscous shear stress $\tau = -\mu \frac{du}{dr}$ across concentric cylindrical fluid shells, the total volumetric flow rate Q is:

$$Q = \frac{\pi r^4 \Delta P}{8 \mu L} \quad \implies \quad R_{hyd} = \frac{\Delta P}{Q} = \frac{8 \mu L}{\pi r^4}$$

Hagen-Poiseuille hydraulic resistance equation highlighting inverse fourth-power radius dependence.

Clinical Impact of the 4th Power Law: Because resistance scales inversely with the fourth power ($R \propto r^{-4}$), a **16% reduction in arteriolar radius halves blood flow** ($0.84^4 \approx 0.50$). Conversely, a 50% stenosis in a coronary artery increases vascular resistance by **16-fold**, producing severe exercise-induced angina and subendocardial ischemia.

3.3. Womersley Number & Pulsatile Flow Regimes

In large conduit vessels (ascending aorta, carotid arteries), blood flow is strongly pulsatile and inertia dominates over viscous forces. The dimensionless **Womersley number** α characterizes the frequency-dependent velocity profile:

$$\alpha = r \sqrt{\frac{\omega \rho}{\mu}} = r \sqrt{\frac{2 \pi f \rho}{\mu}}$$

Where $\alpha > 10$ in human aorta (flat/plug-like velocity profile), and $\alpha < 1$ in arterioles (parabolic Poiseuille flow).

3.4. Reynolds Number & Transition to Turbulence

Turbulent blood flow generates acoustic vibrations heard clinically as murmurs and bruits. The Reynolds number Re determines this transition:

$$Re = \frac{\rho v D}{\mu}$$

Where critical threshold $Re_{crit} \approx 2000 - 2300$ in straight vessels, but drops to ~500 across stenotic, irregular cardiac valves.

3.5. Murray's Law of Minimal Work for Vascular Branching

Cecil D. Murray (1926) established that mammalian vascular branching minimizes total physiological work (viscous dissipation work + metabolic maintenance of blood volume). At every bifurcation of parent vessel radius r_0 into daughter vessels r_1, r_2:

$$r_0^3 = \sum_{i=1}^n r_i^3 = r_1^3 + r_2^3$$

Murray's cubic branching law governing anatomical arborization in coronary and systemic arterial trees.

4. Cardiac Electrophysiology & 12-Lead Electrocardiography

4.1. Sarcolemmal Ionic Gradients & Goldman-Hodgkin-Katz (GHK) Equation

Resting and active cardiomyocyte membrane potentials (V_m) arise from selective ionic permeabilities ($P_K, P_{Na}, P_{Ca}, P_{Cl}$) and asymmetrical trans-sarcolemmal concentration gradients maintained by Na^+/K^+-ATPase pumps:

$$V_m = \frac{RT}{F} \ln \left(\frac{P_{K}[\text{K}^+]_o + P_{Na}[\text{Na}^+]_o + P_{Cl}[\text{Cl}^-]_i}{P_{K}[\text{K}^+]_i + P_{Na}[\text{Na}^+]_i + P_{Cl}[\text{Cl}^-]_o} \right)$$

Goldman-Hodgkin-Katz (GHK) voltage equation for resting and depolarized membrane potential.

4.2. 1D Cable Theory of Myocardial Conduction

Cardiac action potential propagation through gap-junction-coupled syncytial trabeculae obeys the classical cable equation:

$$\lambda^2 \frac{\partial^2 V_m}{\partial x^2} - \tau_m \frac{\partial V_m}{\partial t} - V_m = 0, \quad \text{where } \lambda = \sqrt{\frac{r_m}{r_i + r_o}}, \quad \tau_m = r_m c_m$$

Space constant λ [mm] and membrane time constant τ_m [ms] dictating conduction velocity $\theta \propto \lambda / \tau_m$.

4.3. Einthoven's Triangle & Dipole Field Projection onto 12 Leads

Willem Einthoven (1906) demonstrated that total electrical cardiac activity can be modeled as an instantaneous equivalent dipole vector $\mathbf{D}(t) = [D_x(t), D_y(t), D_z(t)]^T$ in 3D torso space. Any lead potential $V_{lead}(t)$ represents the scalar projection of this cardiac dipole onto the standardized lead vector $\mathbf{L}_{lead}$:

$$V_{lead}(t) = \mathbf{D}(t) \cdot \mathbf{L}_{lead} = D_x L_x + D_y L_y + D_z L_z$$

Fundamental vectorcardiographic lead projection theorem.

Wilson Central Terminal (WCT): Frank N. Wilson (1946) established a virtual zero-potential reference for the precordial leads by connecting the three limb electrodes (Right Arm V_R, Left Arm V_L, Left Leg V_F) through equal $5\,\text{k}\Omega$ resistors:

$$V_{WCT} = \frac{V_R + V_L + V_F}{3} \equiv 0\,\text{mV}$$

Wilson Central Terminal virtual reference node.

Goldberger Augmented Limb Leads: Emanuel Goldberger (1942) removed the connection from the explored limb to the central terminal, augmenting signal voltage by 50% ($1.5\times$):

$$aVR = V_R - \frac{V_L + V_F}{2}, \quad aVL = V_L - \frac{V_R + V_F}{2}, \quad aVF = V_F - \frac{V_R + V_L}{2}$$

Goldberger augmented unipolar limb leads.

4.4. Ischemic Injury Current & Reciprocal ST-Segment Mechanics

During acute transmural myocardial infarction (STEMI), ischemic cardiomyocytes lose intracellular potassium ($[\text{K}^+]_o$ rises), partially depolarizing resting potential from $-90\,\text{mV}$ to $-60\,\text{mV}$ and shortening action potential duration. This creates a continuous systolic injury dipole vector $\Delta \mathbf{D}_{ischemia}$ pointing from normal myocardium toward the ischemic zone:

$$\Delta V_{ST} = \Delta \mathbf{D}_{ischemia} \cdot \mathbf{L}_{lead}$$

Injury current vector producing ST-elevation in facing leads and reciprocal ST-depression in opposite leads.

Example: In Acute Inferior STEMI (Right Coronary Artery occlusion), $\Delta \mathbf{D}_{ischemia}$ points downward and backward ($+100^\circ$). This aligns positively with leads II ($+60^\circ$), III ($+120^\circ$), and aVF ($+90^\circ$) causing marked ST elevation (+3 to +5 mm), while projecting negatively onto high lateral leads I (0°) and aVL (-30°) causing reciprocal ST depression (-2 to -4 mm).

5. Clinical Bioacoustics & Stethoscope Audio Synthesis

5.1. Damped Harmonic Oscillator Model for Valvar Heart Sounds (S1 & S2)

Heart sounds (S1, S2, S3, S4) do not arise from valve leaflets clapping together like doors; rather, they represent the sudden deceleration of blood volume against taut, closed valve curtains, driving the entire cardio-hemic system into damped acoustic oscillation:

$$m \frac{d^2 x}{dt^2} + c \frac{dx}{dt} + k x = F(t) \quad \implies \quad x(t) = A_0 e^{-\zeta \omega_n t} \sin(\omega_d t + \phi)$$

Damped second-order acoustic oscillator where \omega_n = \sqrt{k/m} is natural frequency and \zeta = c / (2\sqrt{km}) is damping ratio.

In Orbit's audio engine, **S1** (Mitral/Tricuspid closure) is synthesized as a low-frequency damped wave (40-70 Hz, duration 100-120 ms, $\zeta \approx 0.18$), whereas **S2** (Aortic/Pulmonic closure) exhibits higher stiffness and lower vibrating mass, yielding higher frequency (80-120 Hz, duration 70-90 ms, $\zeta \approx 0.22$).

5.2. Curle's Aeroacoustic Dipole Radiation Equation for Murmurs

Turbulent flow through stenotic or regurgitant orifices generates pressure fluctuations across solid boundaries (valve leaflets, ventricular walls). N. Curle (1955) extended Lighthill's acoustic analogy to incorporate rigid boundaries, demonstrating that fluctuating surface forces F_i act as acoustic dipoles:

$$p'(\mathbf{x}, t) \approx \frac{1}{4 \pi c_0 r} \int_S \left[\frac{\partial F_i(\mathbf{y}, t')}{\partial t} \right]_{t' = t - r/c_0} \frac{x_i}{r} dS(\mathbf{y})$$

Curle's acoustic dipole radiation equation for murmur generation at fluid-solid interfaces.

Acoustic Power Scaling: Curle's equation dictates that the radiated acoustic sound power W_{sound} scales with the sixth power of jet velocity:

$$W_{sound} \propto \frac{\rho_0 u_{jet}^6 L^2}{c_0^3}$$

Sixth-power acoustic velocity scaling law for cardiac murmurs.

Because velocity across a stenotic valve scales with pressure drop via the Gorlin/Torricelli equation ($u_{jet} = \sqrt{2 \Delta P / \rho}$), even modest elevations in transvalvular gradient ΔP cause exponential surges in high-frequency murmur loudness.

5.3. Verified Hemodynamic Timing & Acoustic Spectral Envelopes

Orbit's Web Audio DSP engine simulates distinct clinical murmurs with authentic spectral shaping and chest-wall formant resonance:

Pathology	Timing in Cycle	Spectral Center	Envelope Profile	Auscultation Site & Radiation
Aortic Stenosis	Systolic ejection (mid-systole)	200 - 550 Hz (Harsh)	Diamond crescendo-decrescendo	2nd R ICS; radiates to carotids
Mitral Regurgitation	Holosystolic / Pansystolic	250 - 650 Hz (Blowing)	Flat rectangular plateau	Apex (mitral); radiates to axilla
Aortic Regurgitation	Early diastole (post-A2)	300 - 750 Hz (High-pitch)	Immediate decrescendo	Erb's point (3rd L ICS) leaning forward
Mitral Stenosis	Mid-diastolic with pre-systolic	60 - 160 Hz (Low rumble)	OS -> low rumble -> atrial kick	Apex in left lateral decubitus (Bell)
Friction Rub	Triphasic (Atrial, Vent, Filling)	350 - 900 Hz (Scratchy)	High-frequency superficial burst	Lower left sternal border held in exp

6. WebGL Shader Mathematics & Mobile Stability Architecture

6.1. Dynamic Shader Discard vs Mesh Allocation Overheads

Rendering 2,234 segmented anatomical structures on mobile Safari/WebKit without memory exhaustion requires eliminating Three.js mesh allocation churn. In Orbit, all segmented structures within an organ system are merged into a single `BufferGeometry`. A per-vertex attribute `partIndex` flags structure identity. In the fragment shader:

$$\text{Fragment Shader: } \quad \mathbf{if}\;(vVisibility < 0.5)\;\mathbf{discard};$$

Early-Z fragment discard rejected at rasterization stage with zero CPU draw call overhead.

6.2. Mobile Jetsam Out-of-Memory (OOM) Prevention Protocol

Mobile WebKit imposes strict per-tab memory ceilings (typically 280MB - 350MB). Concurrent uncompressed ArrayBuffers and unbounded `DecompressionStream` pipelines trigger instant Jetsam kills. Orbit implements four structural guardrails:

1. Bounded Concurrency Queue: Chunk downloads throttled to 2 concurrent pipelines on touch devices (`fetchChunksWithLimit(chunks, 2)`).

2. Instant Geometry Disposal: Temporary `BufferGeometry` instances disposed immediately post-merge.

3. DPR Clamping: `devicePixelRatio` clamped to 1.0 on mobile to avoid 9x fill-rate explosion on 3x Retina displays.

4. DOM Tab Persistence: `` canvas preserved in DOM via CSS `display: none` to eliminate WebGL context recreation churn.

7. Comprehensive Academic Bibliography & Historical Literature

Einthoven, W. (1906): Le télécardiogramme. *Archives Internationales de Physiologie*, 4(2), 132-164. [Foundation of string galvanometer and 3D lead vector projection].

Wiggers, C. J. (1915): *Modern Aspects of the Circulation in Health and Disease*. Lea & Febiger, Philadelphia. [First synthesis of the 7-phase cardiac cycle Wiggers diagram].

Poiseuille, J. L. M. (1840): Recherches expérimentales sur le mouvement des liquides dans les tubes de très petits diamètres. *Comptes Rendus de l'Académie des Sciences*, 11, 961-967. [Laminar flow resistance fourth-power law].

Suga, H., & Sagawa, K. (1974): Instantaneous pressure-volume relationships and their ratio in the excised, supported canine left ventricle. *Circulation Research*, 35(1), 117-126. [Time-varying elastance E(t) model].

Westerhof, N., et al. (1971): Arterial impedance. In *Cardiovascular Fluid Dynamics* (Vol. 1, pp. 119-144). Academic Press. [4-Element Windkessel arterial impedance network].

Murray, C. D. (1926): The physiological principle of minimum work. I. The vascular system and the cost of blood volume. *Proceedings of the National Academy of Sciences*, 12(3), 207-214. [Cubic branching scaling law].

Hodgkin, A. L., & Katz, B. (1949): The effect of sodium ions on the electrical activity of the giant axon of the squid. *The Journal of Physiology*, 108(1), 37-77. [Goldman-Hodgkin-Katz membrane voltage formulation].

Wilson, F. N., et al. (1946): The precordial electrocardiogram. *American Heart Journal*, 32(3), 277-310. [Wilson Central Terminal WCT reference node].

Curle, N. (1955): The influence of solid boundaries upon aerodynamic sound. *Proceedings of the Royal Society of London. Series A*, 231(1187), 505-514. [Aeroacoustic dipole murmur radiation].

Guyton, A. C., & Hall, J. E. (2020): *Textbook of Medical Physiology* (14th ed.). Elsevier, Philadelphia. [Venous return curves, mean systemic filling pressure, and microvascular filtration].

Braunwald, E., et al. (2022): *Braunwald's Heart Disease: A Textbook of Cardiovascular Medicine* (12th ed.). Elsevier. [Clinical auscultation, valve mechanics, and ischemic vector patterns].

Monograph Verification Notice: All mathematical formulas, physical units, constants, and physiological ranges in this document have been computationally verified against the Orbit Virtual Patient Simulator test suite and benchmarked against human clinical telemetry.